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Sex–specific Single Transcript Level Atlas of Vasopressin and its Receptor (AVPR1a) in the Mouse Brain

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eLife Assessment

This work presents a brain-wide atlas of vasopressin (Avp) and vasopressin receptor 1A (Avpr1a) mRNA expression in mouse brains using high-resolution RNAscope in situ hybridization. The single-transcript approach provides precise localization and identifies additional brain regions expressing Avpr1a, creating a **valuable** resource for the field. The revised manuscript is clearer and more impactful, with improved figures, stronger data organization, and enhanced scholarship through added context and citations. Overall, the evidence is **compelling**, and the atlas should be broadly of use to researchers studying vasopressin signaling and related neural circuits.

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Abstract

Vasopressin (AVP), a nonapeptide synthesized predominantly by magnocellular hypothalamic neurons, is conveyed to the posterior pituitary *via* the pituitary stalk, where AVP is secreted into the circulation. Known to regulate blood pressure and water homeostasis, it also modulates diverse social behaviors, such as pair–bonding, social recognition and cognition in mammals including humans. Importantly, AVP modulates social behaviors in a sex–specific manner, perhaps, due to sex differences in the distribution in the brain of AVP and its main receptor AVPR1a. There is a *corpus* of integrative studies for the expression of AVP and AVPR1a in various brain regions, and their functions in modulating central and peripheral actions. In order to purposefully address sexually dimorphic and novel roles of AVP on central and peripheral functions through its AVPR1a, we utilized RNAscope to map *Avp* and *Avpr1a* single transcript expression in the mouse brain. As the most comprehensive atlas of AVP and AVPR1a in the mouse brain, this compendium highlights the importance of newly identified AVP/AVPR1a neuronal nodes that may stimulate further functional studies.

Introduction

Vasopressin (AVP), a nonapeptide synthesized primarily by magnocellular neurons of the paraventricular nucleus (PVH) and supraoptic nucleus (SO) of the hypothalamus, is conveyed along axons of magnocellular neurons *via* the pituitary stalk to the posterior pituitary, where AVP is released into circulation to exert hormonal functions (1, 2). Involved in the regulation of blood pressure and water balance (3, 4), AVP also regulates diverse social behaviors, such as pair–bonding, social recognition and cognition in all mammals, including humans (5–9). It should be noted that similar to OXT, AVP is evolutionarily conserved across invertebrates and vertebrate taxa, differing from OXT by just only two amino acids. Importantly, AVP modulates social behaviors in sex–specific manner, perhaps due to sex differences in AVP and its receptor expression in the brain (10–15).

AVP receptors are G protein–coupled receptors consisting of two major subtypes, the V1 receptor (AVPR1) and V2 receptor (AVPR2) (16). AVPR1 has two subtypes, namely AVPR1a and AVPR1b that mediate the effects of AVP on social behaviors. Avpr1a will be the overarching focus of this study given its abundance and ubiquity in brain regions (9). In fact, differences in AVPR1a genetic variability and expression patterns determine specific social phenotypes (17–21). There is evidence that AVPR1a is involved in maternal care, pair–bonding, behavioral aggression, anxiety, social recognition and social play (6, 9, 19, 22–25). The sex–dependent distribution of Avpr1a across the brain in different species provides a proxy for the distribution of AVP binding, and therefore, provides further evidence for central AVP neural nodes of physiologic relevance.

Blocking AVPR1a inhibits social recognition in the rat, while AVPR1a knockout mice fail to display social recognition (15, 22, 26). These effects of AVP in enhancing social recognition is mediated *via* AVPR1a in the lateral septum (15, 26). Sex differences in Avpr1a binding densities have been described in several brain sites of Wistar rats. Namely, males display higher AVPR1a binding densities in the following forebrain areas: somatosensory and piriform cortex, medial posterior bed nucleus of the stria terminalis (BNST), nucleus of the lateral olfactory tract (LOT), anteroventral thalamic nucleus (VA), tuberal lateral hypothalamus (LH), stigmoid hypothalamus (Stg), and dentate gyrus (DG) (16). In male prairie voles, central AVP infusion facilitates selective aggression associated with pair bond formation and partner preferences, and the Avpr1a antagonist 1-(β-mercapto-β, β-cyclopentamethylene propionic acid) does not seem to inhibit aggression (10). In contrast, central AVP infusion induces a partner preference in female prairie voles (11). Comparative analysis of AVPR1a distribution has revealed higher densities of AVPR1a binding in the ventral pallidum, amygdala, and thalamus of prairie voles than that of meadow or montane voles (27, 28). It has also been reported that AVPR1a antagonism specifically in the ventral pallidum prevents mating–induced partner preferences in male prairie voles (29).

Although osmotically stimulated AVP release with short latency and duration from hypothalamic PVH and SO magnocellular axon terminals occurs in the pituitary (30, 31), it is also released locally from somata and dendrites in the SO with a longer delay responding to osmotic challenge (32, 33). Such local release is likely to facilitate autocrine and/or paracrine regulation of SO–magnocellular neuronal activity and inhibit further systemic AVP output (33, 34). Of note, somatodendritic AVP release in response to direct hypertonic stimulation is attenuated by V1/V2 receptor antagonism, implying that AVP may facilitate its own release by acting on autoreceptors within magnocellular neurons of the SO (35). The importance of autofacilitation to local AVP release may lay in fine–tuned regulation of AVP actions towards physiologic demands. Alternatively, AVP could putatively diffuse over longer distances to bind to adjacent receptors. The relative contribution of AVP autoreceptor subtypes, including AVPR1a, to this phenomenon awaits further clarification.

Current studies implicate posterior pituitary hormones, traditionally thought of as master regulators of a single physiological target, in the control of multiple bodily systems, either directly or *via* their receptors in the brain (23, 36–39). Non–traditional actions of AVP include its ability to affect skeletal homeostasis, wherein it negatively regulates osteoblasts and stimulates osteoclasts. This explains the bone loss that accompanies low blood sodium levels in patients with high AVP levels (40). Furthermore, we have shown that AVP (via AVPR1a) and oxytocin (via OXTR) have opposing skeletal actions—effects that may relate to the pathogenesis of bone loss in chronic hyponatremia, and pregnancy and lactation, respectively (40–43).

Detecting specific AVPR1a and AVPR1b in the brain has had limitations for a long time due to the availability of only nonselective radioligands, such as tritium labeled (^{3}H) AVP ligands, which bind to both receptors (44). Although there is a large body of integrative studies for the expression of AVP and AVPR1a in various brain regions, and their functions in regulating central and peripheral actions, there remains the need for detailed, sex–specific mapping of the AVP/AVPR1a neuronal nodes in the brain. We utilized RNAscope—a technology that detects single RNA transcripts—to create a comprehensive atlas of AVP and AVPR1a in the mouse brain. It may seem somewhat remarkable that newly discovered brain areas for receptors of such evolutionarily conserved and well–studied hormones AVP and OXT are currently emerging, with inferences to novel functions.

We believe that this atlas of AVP and its AVPR1a in concrete brain sites at a single transcript level should provide a resource to neuroscientists to deepen our understanding of classical and novel central and peripheral functions of AVP by interrogating AVPR1a site–specifically.

Results

AVP receptors are G protein-coupled receptors consisting of two major subtypes, the AVPR1 and AVPR2 (16). In turn, AVPR1 is divided into AVPR1a and AVPR1b receptor subtypes that mediate the effects of AVP in the brain on social behaviors. In this study, we have provided not only a distribution mapping of AVP and AVPR1a in the brain, but also assessed sex differences in their expression by RNAscope. Allowing the detection of single transcripts, RNAscope uses ~20 pairs of transcript–specific double *Z*–probes to hybridize 10–µm–thick whole brain sections. Preamplifiers first hybridize to the ~28–bp binding site formed by each double *Z*–probe; amplifiers then bind to the multiple binding sites on each preamplifier; and finally, labeled probes containing a chromogenic enzyme bind to multiple sites of each amplifier.

RNAscope data were quantified on every tenth section of the whole brains from coded 3 female and 3 male mice. For simplicity and clarity in the graphs, a scatter plot has been shown for 3 nuclei, sub–nuclei and regions with the highest AVP and AVPR1a transcript densities as well as their absolute transcript counts. Each section was viewed and analyzed using CaseViewer 2.4 (3DHISTECH, Budapest, Hungary) and QuPath v.0.2.3 (University of Edinburgh, UK). The *Atlas for the Mouse Brain in Stereotaxic Coordinates* (45) was used to identify every nucleus, sub–nucleus or region, which was followed by manual counting of *Avp* and *Avpr1a* transcripts by two independent observers (V.R. and A.G.) in every tenth section using a tag feature. Receptor density was calculated by dividing the absolute receptor number by the total area (µm^2, ImageJ) of every nucleus, sub–nucleus or region. The highest *Avp* and *Avpr1a* values in the brain regions are presented as means ± SE and compared with those of the opposite sex. Photomicrographs were prepared using Photoshop CS5 (Adobe Systems) only to adjust brightness, contrast and sharpness, to remove artifacts (*i.e.*, obscuring bubbles), and to make composite plates.

RNAscope revealed *Avp* expression in the hypothalamus, forebrain, hippocampus and cortex of both female and male mice (Fig. 1A), however, *Avp* expression in the 3rd ventricular region and thalamus was found only in the female mouse (Fig. 1B). Whereas the numbers of *Avp*–expressing cells were greater in females compared with males in the hypothalamus (607 vs. 471), 3rd ventricular region (7 vs. 0) and thalamus (2 vs. 0), those numbers were lower in the hippocampus (58 vs. 118), forebrain (50 vs. 77) and cortex (5 vs. 6) [see Appendix 1 for Glossary and Supplementary Fig. 1 for raw count graphs].

The highest *Avp* transcript densities were detected in following brain nuclei, sub– nuclei and regions of female and male mice, respectively: ventricular region—3V only for females, hypothalamus—SChVL and PaDC, forebrain—MPOM and VLPO, hippocampus—Py and GrDG, thalamus—ic only for females and cerebral cortex—Pir for both (Fig. 1B). Representative micrographs of some of the hypothalamic and forebrain regions with highest *Avp* expression are shown in Fig. 1C.

We found that *Avpr1a* transcript expression in several brain nuclei, for example, the MS of the forebrain, medullary IOBe, IRt, LRt (Supplementary Fig. 2B), displayed individual patterns of expression *vs.* a more ubiquitous and even expression noted in most of the other brain areas, suggesting brain–site–specific functional diversity and context–selective regulation of *Avpr1a*–mediated signaling. We report the expression of the *Avpr1a* in 398 and 375 brain nuclei, sub–nuclei and regions of the female and male mice, respectively. *Avpr1a* transcripts were detected bilaterally, with no apparent ipsilateral domination. Probe specificity was established by a positive signal in renal tubules of the kidney (positive control) with an absent signal in the FrA of the frontal cortex (negative control) (Fig. 2A). Representative micrographs of sex–specific medullary CC and hypothalamic Arc with highest *Avpr1a* expression also are shown in Fig. 2A.

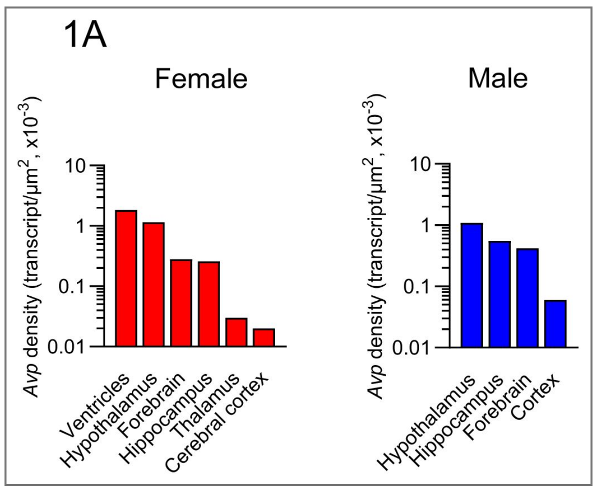
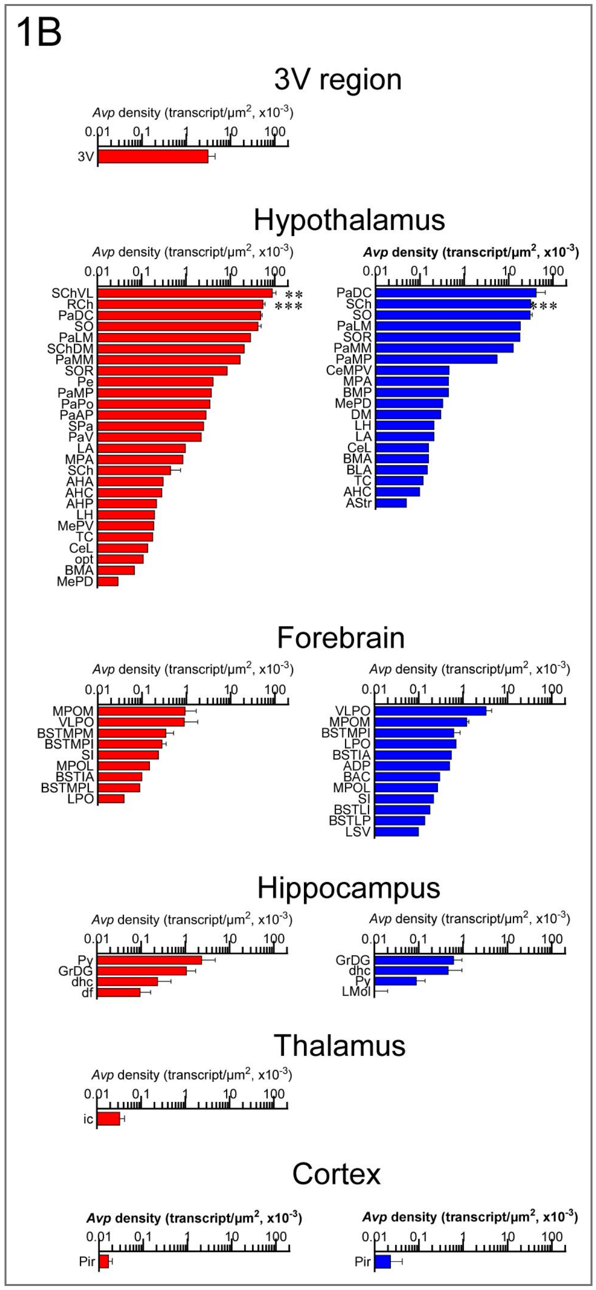


Figure 1. Sex–specific *Avp* transcript density in the brain.

(**A**) *Avp* transcript density in main brain regions detected by RNAscope. Female 3V region and hypothalamus and male hypothalamus had the highest *Avp* transcript density. (**B**) Representative micrographs of some of the hypothalamic and forebrain regions with highest *Avp* expression. Scale bar: 20 µm. (**C**) *Avp* transcript density in nuclei, sub– nuclei and regions of the hypothalamus and forebrain. (**D**) Novel *Avp* transcripts found in nuclei, sub–nuclei and regions of the hippocampus, thalamus and cerebral cortex. Scale bar: 10 µm. Note, that *Avp* transcripts were found only in the female 3V ependymal layer and thalamus. *N*=3 mice *per* sex.



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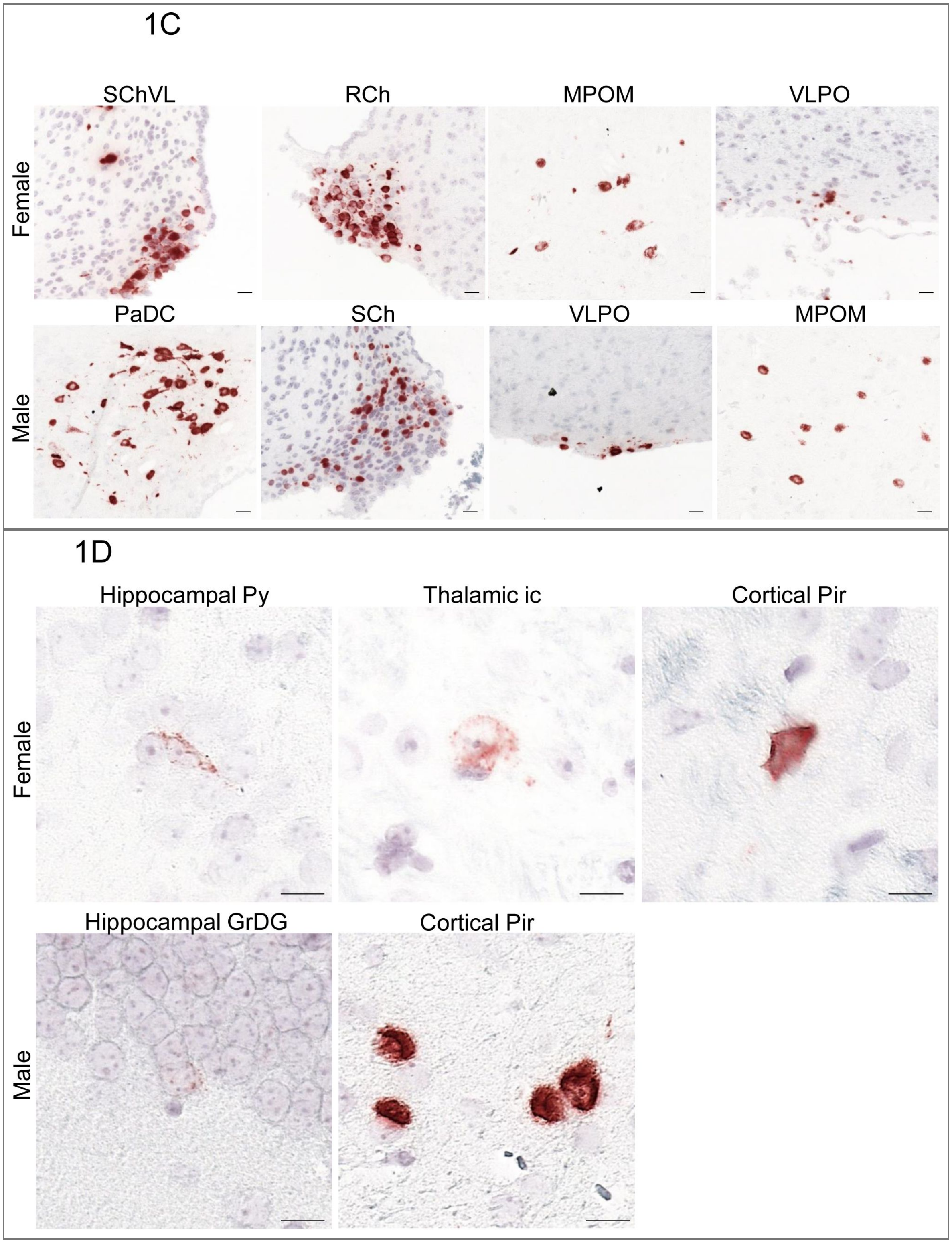

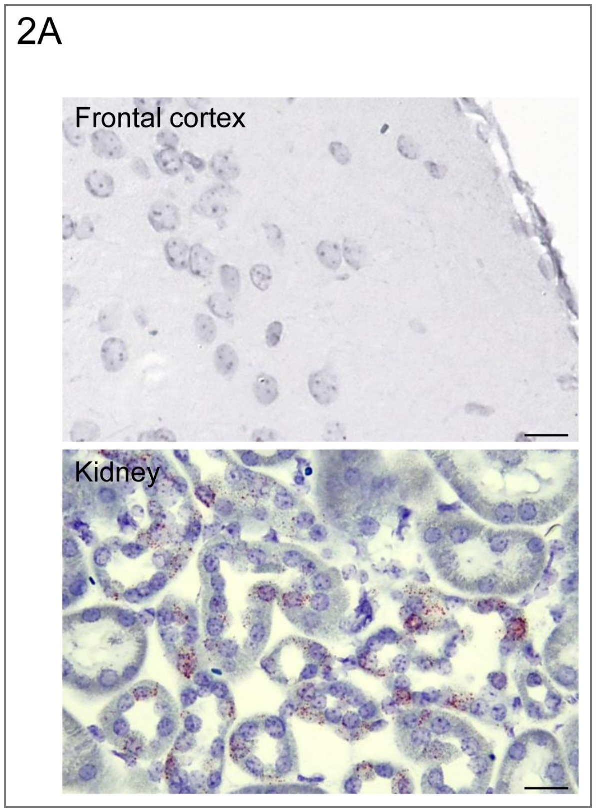

Figure 2. **Sex–specific *Avpr1a* transcript density in the brain.**

(**A**) Probe specificity was established by a positive signal in renal tubules of the kidney (positive control) with an absent signal in the frontal association cortex (FrA) (negative control). Also shown are representative micrographs of medullary central canal (CC) and hypothalamic arcuate nucleus (Arc). (**B**) *Avpr1a* transcript density in main brain regions detected by RNAscope. (**C**) *Avpr1a* transcript density in nuclei, sub–nuclei and regions of the ventricular regions, hypothalamus, medulla, midbrain and pons, forebrain, olfactory bulb, hippocampus, thalamus, cortex and cerebellum. *N*=3 mice *per* sex. Scale bar: 25 µm (controls) and 20 µm in (A).

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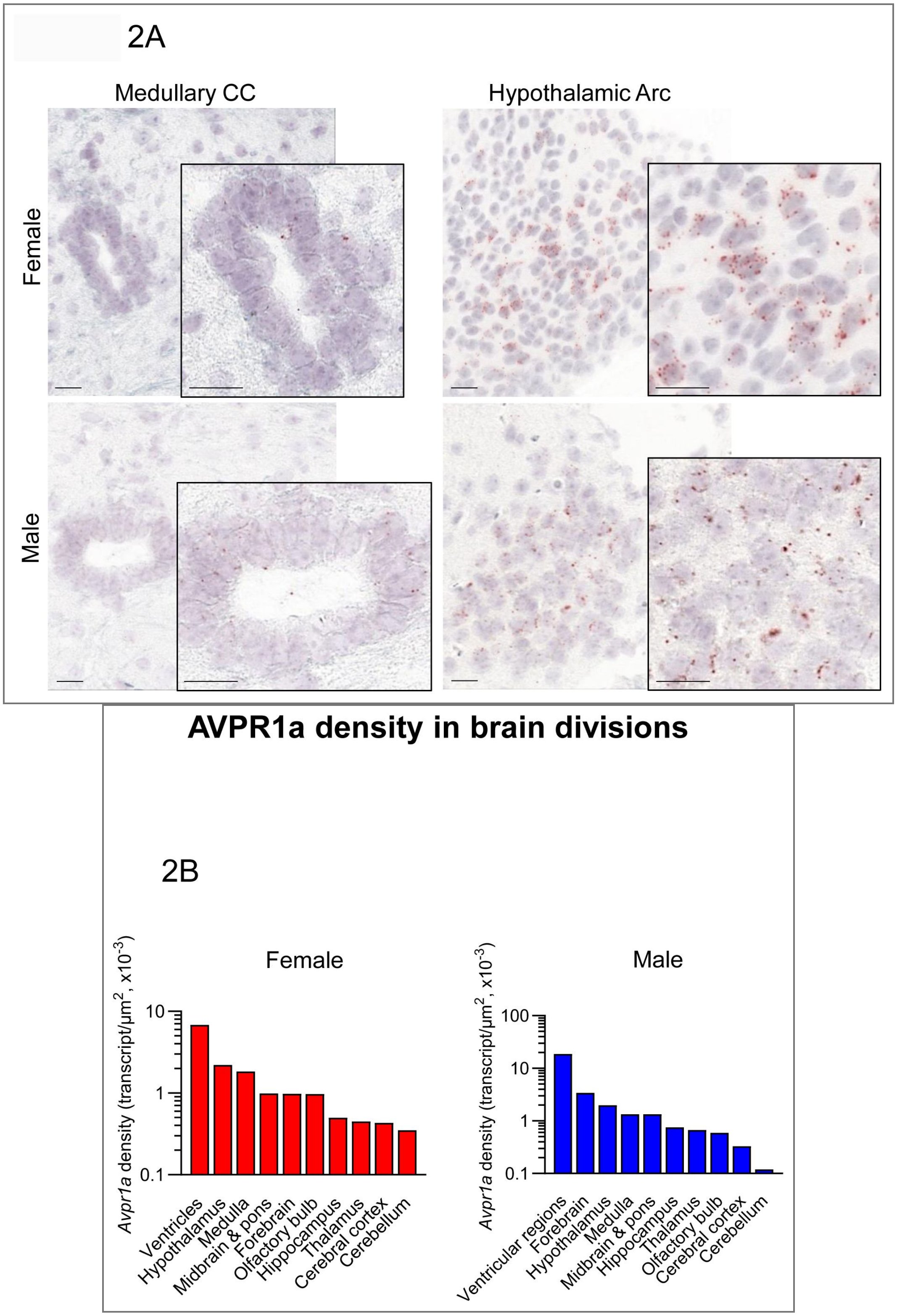

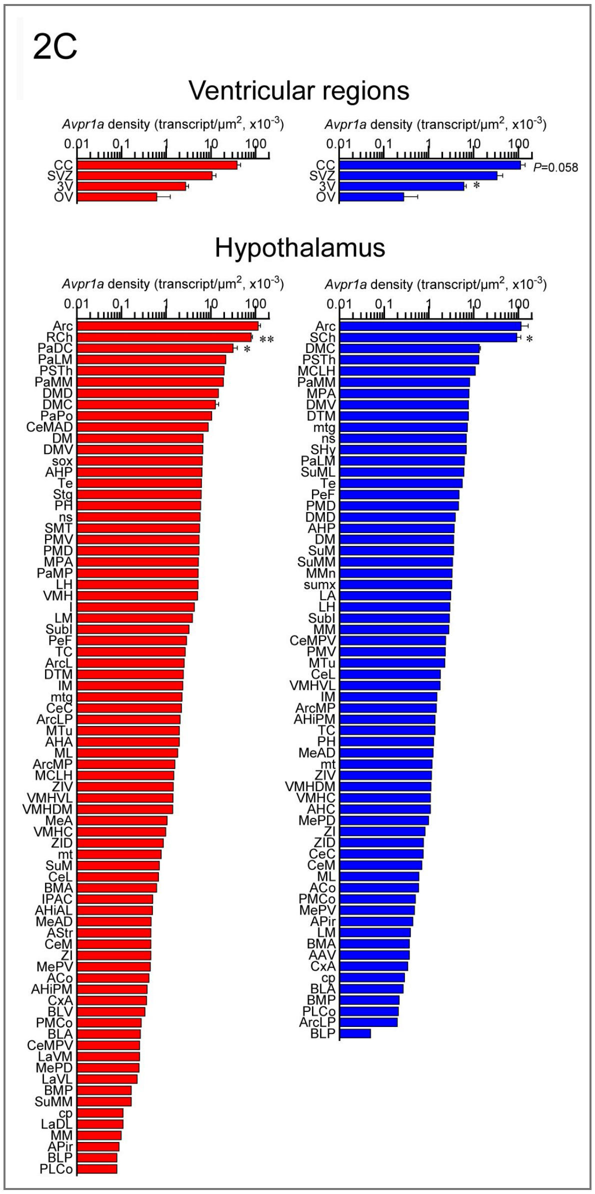

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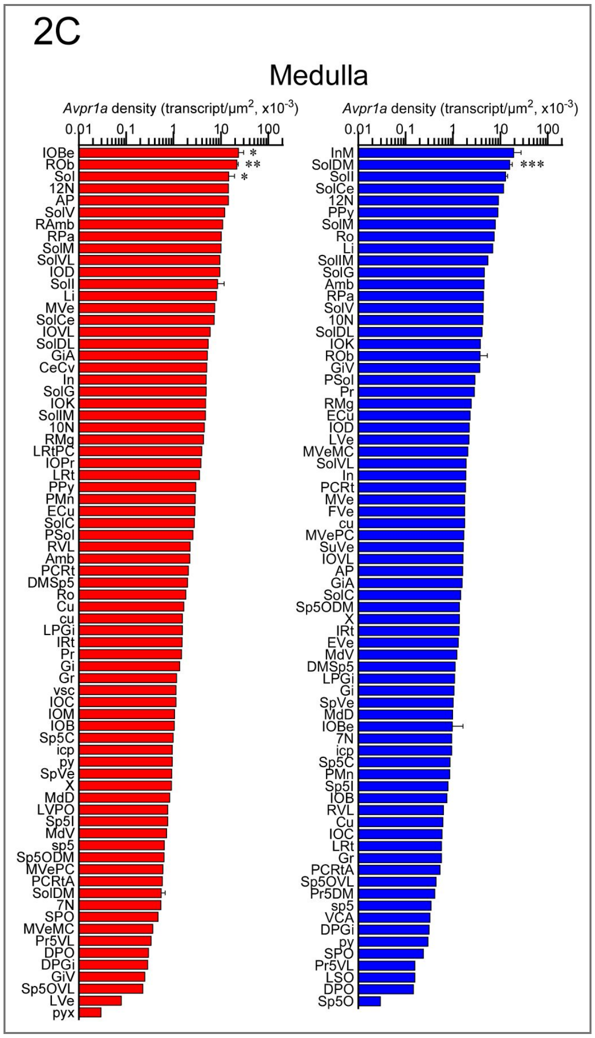

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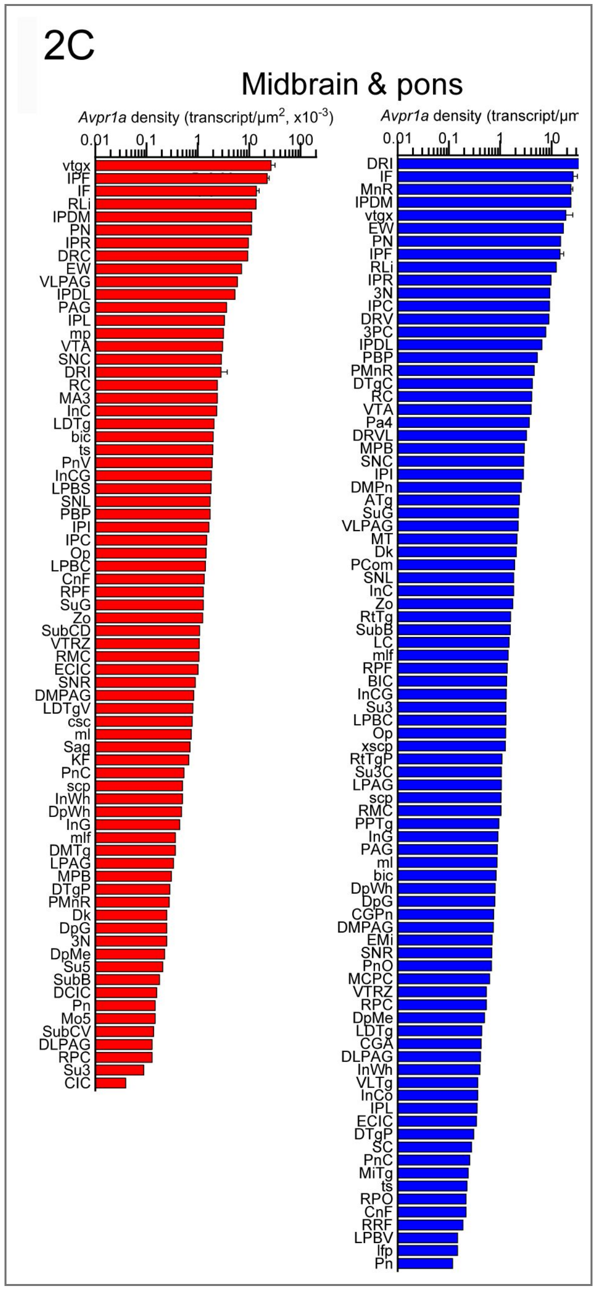

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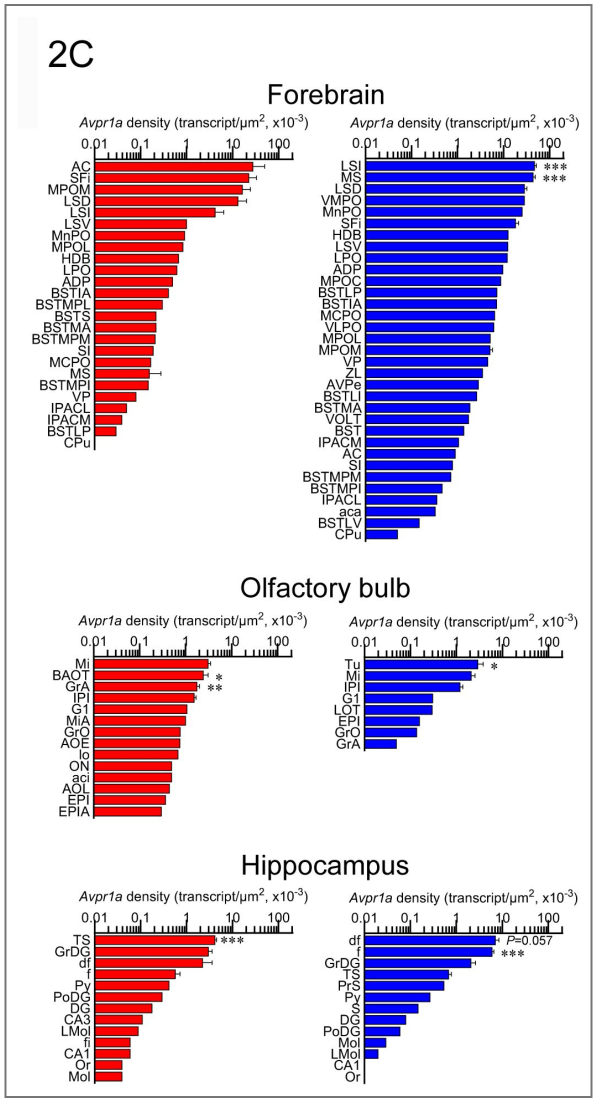

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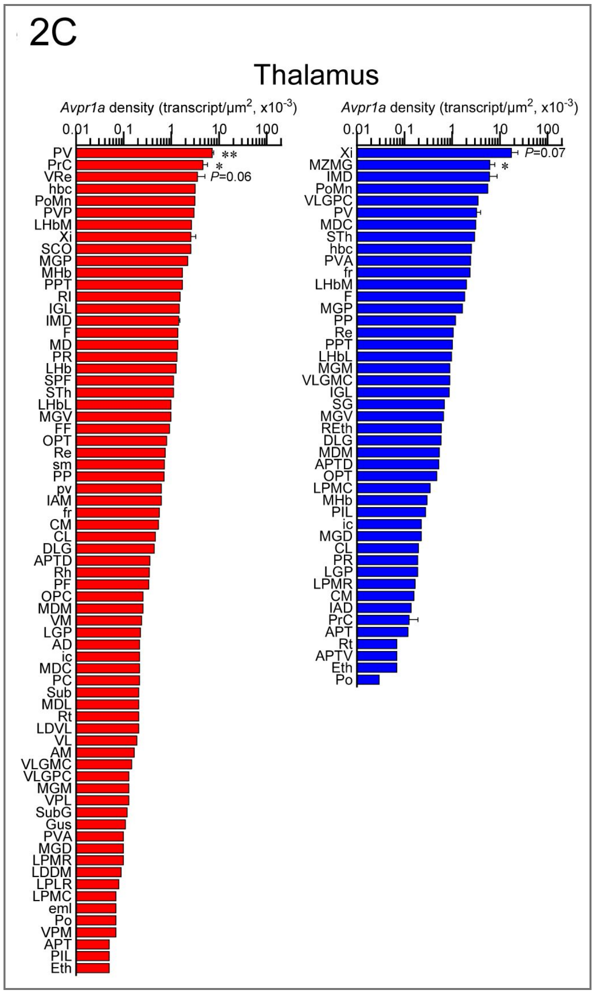

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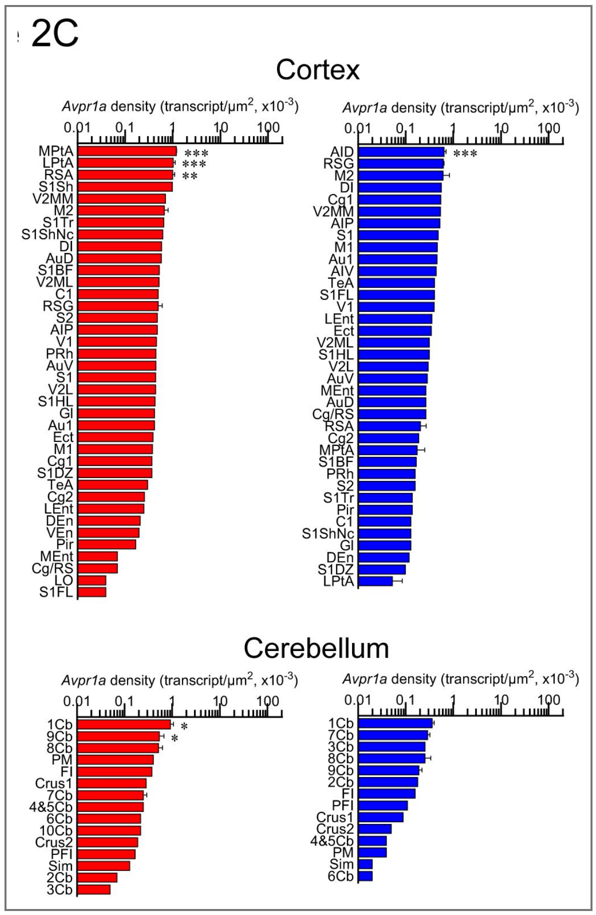


Figure 2. (continued)

In the female, total *Avpr1a* transcript numbers were the highest in the medulla followed, in descending order, by the hypothalamus, cortex, midbrain and pons, forebrain, thalamus, cerebellum, hippocampus, olfactory bulb and ventricular regions (Supplementary Fig. 2A). In the male, *Avpr1a* transcripts were the highest in the forebrain, followed by the medulla, midbrain and pons, hypothalamus, cortex, hippocampus, thalamus, olfactory bulb, cerebellum and ventricular regions (Supplementary Fig. 2).

Using the RNAscope dataset we further calculated *Avpr1a* density in all brain divisions (Fig. 2B), nuclei, sub–nuclei and regions (Fig. 2C). Highest *Avpr1a* densities in the female and male mice, respectively, were noted in nuclei, sub–nuclei and regions as follows: ventricular regions—CC ependymal region for both with 2.92–fold greater in the male; hypothalamus—Arc for both; medulla—IOBe and InM; midbrain and pons— vtgx and DRI; forebrain—AC and LSI; olfactory bulb—Mi and Tu; hippocampus—TS and df; thalamus—PV and Xi; cortex—MPtA and AID and cerebellum—1Cb for both with 2.58-fold greater in the female (Fig. 2C).

In addition, RNAscope analysis revealed that various brain nuclei, sub–nuclei and regions in both female and male mice co–localized *Avp* and *Avpr1a* transcripts. *Avpr1a* to *Avp* ratios within the same brain nucleus, sub–nucleus and region in both sexes are demonstrated in Figs. 3A, B. Finally, RNAscope showed *Avp* and *Avpr1a* expression in the posterior pituitary lobe with *Avp* and *Avpr1a* transcript densities that were higher in male compared with female mice, however, without statistical significance (Fig. 4).

Discussion

Here, we attempted to integrate previous information on sex–specific AVP and its AVPR1a expression in the murine brain. AVPR1a is the most abundant and wide spread receptor in the brain (9) that plays a dominant role in regulating behavior. In addition, we focused towards paradigm–shifting non–traditional roles of central AVP signaling in light of newly discovered AVPR1a. We report AVP expression in 41 female and 13 male brain nuclei, sub–nuclei and regions. Moreover, we identified abundant AVPR1a expression in 398 female and 375 male brain nuclei, sub–nuclei and regions. Therefore, this report is the most exhaustive atlas of brain *Avp* and *Avpr1a* expression at the single transcript level.

It has been reported that AVP synthesis and AVP fiber projections are sexually dimorphic in specific brain sites [for review, see: (16)]. To our knowledge, the first discovery of the sexually dimorphic nature of AVP in the rat brain was made by De Vries *et al.* (46). That is, males displayed more AVP–immunoreactive fibers in the lateral septum and lateral habenular nucleus over females (46). Surprisingly, sex differences in AVP fiber density in the LS and medial amygdala (MeA) originate from the BNST, given only lesions to the BNST, but not the PVH, result in decreased AVP fiber density in the LS (47, 48). In adult rats, AVP fiber density from the BNST and MeA are dependent on circulating gonadal hormones, as gonadectomy eliminates AVP expression and hormone replacement restores AVP fiber network (49, 50). Nonetheless, gonadal hormones appear to only partially explain sex difference in AVP expression because both females and males, exposed to a similar gonadal steroid hormone regime, still differ sexually (51, 52).

Magnocellular neurons of the PVN, SO and SCh of the hypothalamus are the predominant source of AVP synthesis. Hypothalamic AVP synthesis of most rodent species is similar between males and females in the PVH and SO of mice (53, 54); PVH, SO and SCh of voles (55, 56); PVH, MPOA, LH and AHA of Mongolian gerbils and Chinese striped hamsters (57) [for review, see: (16)]. In concordance with these reports, we also find no sex difference in *Avp* synthesis, as evidenced by similar numbers of *Avp*–expressing neurons, in the PVH and SO of mice; however, we did note sex differences in *Avp* expression density in specific hypothalamic nuclei. Notably, *Avp* expression density was higher in several PVH (PaLM and PaMM) and suprachiasmatic (SChVL and RCh) subnuclei of female compared to male mice. No sex differences in SO–*Avp* expression density were found.

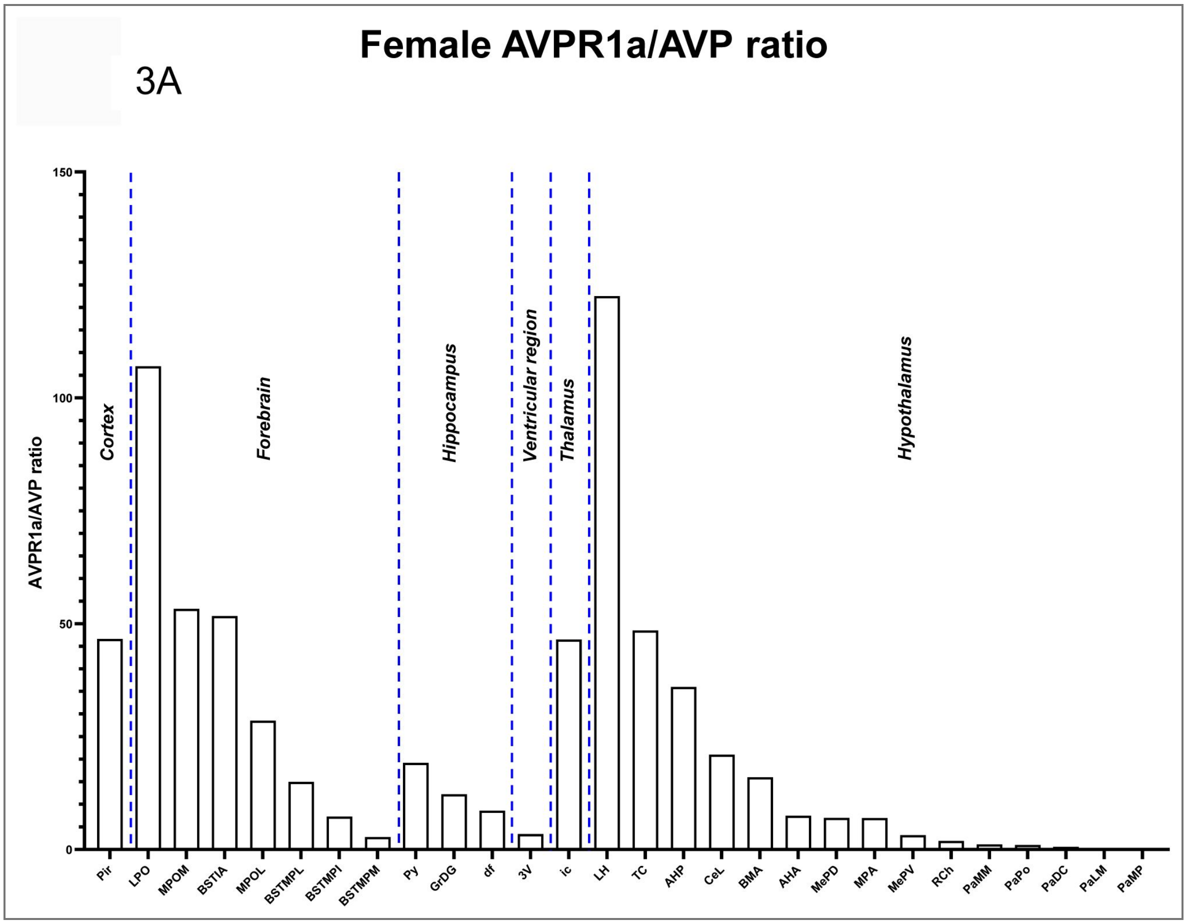


Figure 3. ***Avp*** **and** ***Avpr1a*** **co-localization in the brain.**

We found that various nuclei and sub–nuclei exhibited *Avp* and *Avpr1a* co-localization in the brain of both sexes. (**A**) Female and (**B**) male *Avpr1a* to *Avp* ratios in the cortex, forebrain, hippocampus, 3rd ventricular region, thalamus and hypothalamus.

Figure 3. (continued)

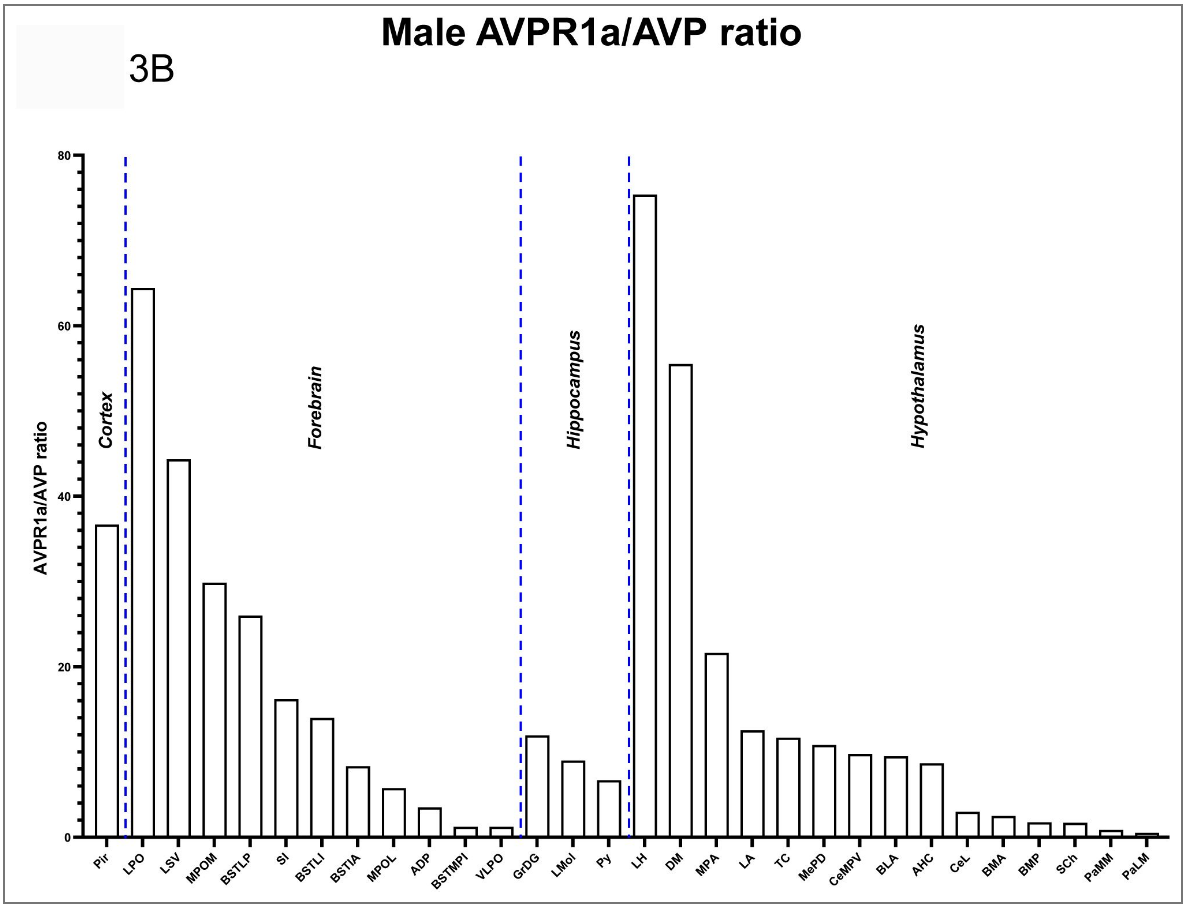
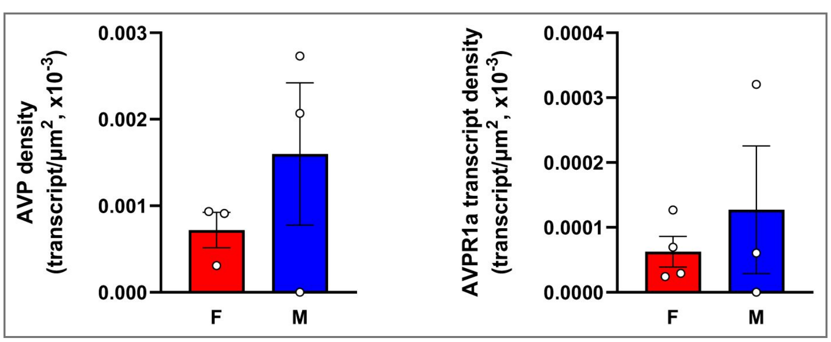


Figure 4. ***Avp* and *Avpr1a* transcript densities in the pituitary gland.**

RNAscope revealed *Avp* and *Avpr1a* expression in the posterior pituitary lobe with *Avp* and *Avpr1a* transcript densities that were higher in male compared with female mice. . *N*=3 mice *per* sex.



Furthermore, *Avpr1a* transcript density was highest in the arcuate nucleus (Arc) and retrochiasmatic sub–nucleus (RCh) in female compared with Arc and suprachiasmatic nucleus (SCh) of male mice. *Avpr1a* expression in the Arc has previously been reported (58), however, its role was unclear until a recent report demonstrating a critical involvement of Arc–NPY in the regulation of fluid homeostasis and the induction of salt water–induced hypertension through AVP modulation in the SO (59); the latter receives direct projections from the Arc (60). In contrast, it is plausible, but, by no means proven, that PVH– and/or SO–AVP may modulate Arc anorexigenic neurons to inhibit ingestive behavior, which has previously been shown with PVH oxytocinergic neurons (61). Indeed, increasing evidence suggests that AVP reduces feeding in mammals (62, 63).

It has been reported that in the rat, AVP is an important output of the SCh targeting AVP cells in other hypothalamic areas—its release into the CSF peaks in the early morning and declines later in a day (64). Specifically, SCh–AVP secreted during late sleep activates osmosensory afferents to AVP neurons in the SO and organum vasculosum of the lamina terminalis (65, 66). Similar to rodents, studies in humans also determined that the main AVP projections from the SCh target the anteroventral hypothalamic area, sub–PVH as well as ventral parts of the PVH and DMH—a remarkable evolutionary conservation of SCh innervation from rodent to human (67, 68).

The fact that the SCh is another brain nucleus with high AVP and AVPR1a expression density (greater in males vs. females), accentuates an important role of SCh–AVP in circadian rhythmicity, notably impacting neuroendocrine day/night rhythms, feeding timing, period, precision, and synchronization of SCh neurons (64, 69, 70).

In the hindbrain, the highest *Avpr1a* transcript density was noted in the inferior olive, beta subnucleus (IOBe) of female and intermedius nucleus (InM) of male mice. It has been reported that AVP fibers are apparent in the hindbrain, such as the parabrachial nucleus, locus coeruleus, and near inferior olive nuclei (71). In this regard, *Avpr1a* mRNA expression has been noted in the inferior olive (58). Given this nucleus has been implicated in various functions including learning and timing of movements, it is possible that AVPR1a in the inferior olive may be activated by the paracrine release of AVP from distant nuclei, such as the SCh, to control motor learning and timing. Alternatively, AVPR1a in the inferior olive may respond to other ligands (for example, OXT) found in nearby regions (72). The role of AVPR1 in the InM of male mice is less clear, but because the InM sends monosynaptic projections to the NTS (73) that is essential for blood pressure control by AVP and receiving information from the cardiovascular receptors (74), a possible coordinated control by hindbrain AVP of blood pressure and cardiovascular function.

Although the midbrain, pons and forebrain displayed less abundant *Avpr1a* transcript density, they revealed further sex differences. In the midbrain and pons, the highest *Avpr1a* density was in the ventral tegmental decussation (vtgx) in females and dorsal raphe nucleus, interfascicular part (DRI) in males. AVP and OXT in the ventral tegmental area are known to regulate social interactions with rewarding properties.

Indeed, humans, as inherently social beings, show a strong inclination to affiliate and share their emotions with each other (75, 76). Sex differences in ventral tegmental AVPR1a make biological sense, as social interaction of females, specifically, with pups and, generally, with counterparts throughout their lives, have rewarding properties fundamental to maternal behavior and survival. Modulation of AVPR1a in the dorsal raphe nucleus has also been linked to social and emotional behaviors (16, 77, 78). Sexual dimorphism in AVP innervation of and AVPR1a expression in the DRI appears to imprint dimorphic social behaviors. That is, AVPR1a blockade in the lateral habenular nucleus (LHb) of males, but not females who have lesser AVP innervation of the LHb and dorsal raphe nucleus, results in reduced urine marking to unfamiliar males and ultrasonic vocalizations to unfamiliar, sexually receptive females, whereas AVPR1a blockade in the dorsal raphe nucleus of only males reduces urine marking to unfamiliar males (77).

In the forebrain, both sexes displayed high *Avpr1a* transcript density in septal nuclei. The highest *Avpr1a* density was in the anterior commissural nucleus (AC) within the septal nuclei of females, and lateral (LSI) and medial (MS) septal nuclei of males. It is not surprising that *Avpr1a* transcript density was significantly greater in septal nuclei of males than females given in many rodent species males have more AVP– immunoreactive fibers in the lateral septum (46). Notably, the effects of AVP on social recognition is mediated *via* AVPR1a in the lateral septum (15, 26). For example, AVPR1a blockade inhibits social recognition in the rat, while AVPR1a knockout mice fail to display social recognition (15, 22, 26).

Despite *Avpr1a* transcript densities in other brain divisions were significantly lower, sexually dimorphic differences are worth mentioning here. In the olfactory bulb, females had the high *Avpr1a* density in the mitral cell layer (Mi), whereas males had high receptor expression in the olfactory tubercle (Tu). A population of AVP neurons in the olfactory bulb of the rat that play a role in social recognition via AVPR1a has been reported (79). Silencing the AVPR1a by siRNA impairs habituation/dishabituation to juvenile cues, but not to volatile odors (79). Of note, AVP is a retrograde signal that filters activation of the Mi cells in the ewe, likely through presynaptic modulation of norepinephrine or acetylcholine. The secretion of both transmitters is stimulated by AVP in the olfactory bulb (79, 80). The functional relevance of AVP signaling *via* AVPR1a activation in the Tu requires additional studies. There is, however, evidence in the rat that AVP *via* AVPR1a has, at least, an indirect impact on Tu function, as seen by a reduction in activation responding to a noxious odor of butyric acid, when the AVPR1a is blocked (81). The presence of AVPR1a in the hippocampus, thalamus, cortex, and ventricular regions are consistent with the reported effects of AVP on memory (82), emotional and reward–motivated behavior (83), blood pressure (84), blood flow and CSF production (85). Functional roles of many other nuclei shown here to express AVPR1a and not mentioned in this report are much less clear. The importance of revealing novel AVP–triggered functions by interrogating AVPR1a site–specifically, will require further investigations.

Collectively, our results provide compelling evidence of distinct and novel AVP/AVPR1a neuronal nodes in the brain. While studies on central AVP signaling and its control of blood pressure, water balance and diverse social behaviors in mammals, occupy the vast majority of the literature (3–9), we expect that this comprehensive compendium of sex–specific AVP/AVPR1a expression in the brain will deepen our understanding of the functional and neuroanatomical basis underlying old and new paradigm–shifting functions of central AVP signaling. As appears to be the case for most brain areas, the original discovery of function tends to become dogma thereby leading to an oversimplification of multiple functions of those brain areas as they interact in circuits. Finally, the approach of direct mapping of receptor expression in the brain and periphery provides the groundwork for greater discernment of new functional arrangements of ancient pituitary glycoprotein hormones and nonapeptides, such as AVP and OXT, and provide helpful pointers towards improving pharmacological interventions in disease.

Methods

Mice

Adult mice (~3 to 4–month–old) were housed in a 12 h:12 h light : dark cycle at 22 ± 2 °C with *ad libitum* access to water and regular chow. All procedures were approved by the Mount Sinai Institutional Animal Care and Use Committee and are in accordance with Public Health Service and United States Department of Agriculture guidelines.

RNAscope

For RNAscope, mice were anesthetized with isoflurane (2-3% in oxygen; Baxter Healthcare, Deerfield, IL) and transcardially perfused with 0.9% heparinized saline followed by 10% Neutral Buffered Formalin (NBF). Brains were promptly extracted, post-fixed in 10% NBF for 24 h, dehydrated, and paraffin–embedded. Coronal sections were cut at 5 µm, with every tenth section mounted onto ~60 slides with 3 sections on each slide. This method allows to cover the entire brain and eliminate the likelihood of counting the same transcript twice. Sections were air-dried overnight at room temperature and stored at 4 °C until required.

Detection of mouse *Avp* and *Avpr* (*Avpr1a*) was performed separately on paraffin sections using Advanced Cell Diagnostics (ACD) RNAscope 2.5 LS Reagent Kit (#322100) and two RNAscope 2.5 LS probes, namely Mm-AVP-O1 (#472268) and Mm-AVPR1a (#418068). The kidney and prefrontal cortex served as positive and negative controls for AVPR1a, respectively. As with AVP, magnocellular cells of the PVH and SON served as positive controls while the brain from AVP–knockout mouse served a negative control.

Slides were baked at 60 °C for 1 h, deparaffinized, incubated with hydrogen peroxide for 10 min at room temperature, pretreated with Target Retrieval Reagent (#322001) for 20 min at 100 °C and with Protease III for 30 min at 40 °C. Probe hybridization and signal amplification were performed as per the manufacturer's instructions for chromogenic assays.

Following RNAscope assay, the slides were scanned at ×20 magnification and the digital image analysis was successfully validated using the CaseViewer 2.4 (3DHISTECH) software. The same software was employed to capture and prepare images for the figures in the article. Images of control tissues were taken using microscope Leica DM 1000 LED. Detection of *Avp*- and *Avpr1a*–positive cells was also performed using the QuPath-0.2.3 (University of Edinburgh, UK) software. *The Atlas for the Mouse Brain in Stereotaxic Coordinates* (45) was utilized to identify and manually map every nucleus, sub–nucleus or region using drawing features of the QuPath-0.2.3 software in every tenth brain section. This was followed by exhaustive counting of *Avp* and *Avpr1a* transcripts using a tag feature. *Avp* and *Avpr1a* transcript density was calculated by dividing the absolute numbers by the total area (μm^2, ImageJ) of every nucleus, sub–nucleus or region. Photomicrographs were prepared using Photoshop CS5 (Adobe Systems) only to adjust brightness, contrast and sharpness, to remove artifacts (*i.e.,* obscuring bubbles), and to make composite plates.

Quantitation, Validation and Statistical Analysis

Data were analyzed by Student's *t*-test and two-way analysis of variance (ANOVA) followed by Tukey's multiple comparisons tests using GraphPad Prism 10.2.2 version (La Jolla, CA). Significance was set at $P < 0.05$. *P* values are shown.

Data availability

Yes, I have a dataset to upload to Dryad.

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Additional files

Supplemental Figure 1 Sex–specific *Avp* transcript numbers in the brain. (**A**) Total numbers of *Avp*–positive cells in main brain regions detected by RNAscope. (**B**) Total numbers of *Avp*–positive cells in nuclei, sub–nuclei and regions of the hypothalamus, hippocampus, forebrain and cortex. Note, that *Avp*–positive cells were found only in the female 3V ependymal layer and thalamus. Also shown are representative RNAscope micrographs of hypothalamic regions with highest *Avp* expression; *i.e.*, supraotpic nucleus (SO), paraventricular hypothalamic nucleus, lateral magnocellular part (PaLM), retrochiasmatic area (RCh), supraoptic nucleus, retrochiasmatic part (SOR) and suprachiasmatic nucleus (SCh). . *N*=3 mice *per* sex. Scale bar: 20 µm.

Supplemental Figure 2 Sex–specific *Avpr1a* transcript numbers in the brain. (**A**) Tolal *Avpr1a* transcript numbers in main brain regions detected by RNAscope. (**B**) *Avpr1a* transcript density in nuclei, sub–nuclei and regions of the medulla, hypothalamus, cortex, midbrain and pons, forebrain, thalamus, cerebellum, hippocampus, olfactory bulb and ventricular regions. Also shown are representative RNAscope micrographs of medullary regions with highest *Avpr1a* expression; *i.e.*, hypoglossal nucleus (12N), intermediate reticular nucleus (IRt), lateral reticular nucleus (LRt) and medullary reticular nucleus, ventral part (MdV). . *N*=3 mice *per* sex. Scale bar: 20 μm.

Appendix 1

Glossary of the brain nuclei, sub–nuclei and regions.

Olfactory bulb	
aci	anterior commissure, intrabulbar part
AOE	anterior olfactory nucleus, external part
AOL	anterior olfactory nucleus, lateral part
BAOT	bed nucleus of the accessory olfactory tract
EPI	external plexiform layer of the olfactory bulb
EPIA	external plexiform layer of the accessory olfactory bulb
G1	glomerular layer of the olfactory bulb
GrA	granule cell layer of the accessory olfactory bulb
GrO	granular cell layer of the olfactory bulb
IPI	interpeduncular nucleus, intermediate subnucleus
lo	lateral olfactory tract
LOT	nucleus of the lateral olfactory tract
Mi	mitral cell layer of the olfactory bulb
MiA	mitral cell layer of the accessory olfactory bulb
ON	olfactory nerve layer
Tu	olfactory tubercle
Cerebral cortex	
AID	agranular insular cortex, dorsal part
AIP	agranular insular cortex, posterior part
AIV	agranular insular cortex, ventral part
Au1	primary auditory cortex
AuD	secondary auditory cortex, dorsal area
AuV	secondary auditory cortex, ventral area
Cg/RS	cingular/retrosplenial cortex
Cg1	cingulate cortex, area 1
Cg2	cingulate cortex, area 2
C1	C1 adrenaline cells
DEn	dorsal endopiriform nucleus
DI	dysgranular insular cortex
Ect	ectorhinal cortex
GI	granular insular cortex
LEnt	lateral entorhinal cortex
LO	lateral orbital cortex
LPtA	lateral parietal association cortex
M1	primary motor cortex
M2	secondary motor cortex
MEnt	medial entorhinal cortex
MPtA	medial parietal association cortex
Pir	piriform cortex
PRh	perirhinal cortex
RSA	retrosplenial agranular cortex
RSG	retrosplenial granular cortex
S1	primary somatosensory cortex
S1BF	primary somatosensory cortex, barrel field
S1DZ	primary somatosensory cortex, dysgranular region
S1FL	primary somatosensory cortex, forelimb region
S1HL	primary somatosensory cortex, hindlimb region
S1Sh	primary somatosensory cortex, shoulder region
S1ShNc	primary somatosensory cortex, shoulder/neck region
S1Tr	primary somatosensory cortex, trunk region

S2	secondary somatosensory cortex
TeA	temporal association cortex
V1	primary visual cortex
V2L	secondary visual cortex, lateral area
V2ML	secondary visual cortex, mediolateral area
V2MM	secondary visual cortex, mediomedial area
VEn	ventral endopiriform nucleus
Forebrain	
AC	anterior commissural nucleus
aca	anterior commissure, anterior part
ADP	anterodorsal preoptic nucleus
AVPe	anteroventral periventricular nucleus
BAC	bed nucleus of the anterior commissure
BST	bed nucleus of the stria terminalis
BSTIA	bed nucleus of the stria terminalis, intraamygdaloid division
BSTLI	bed nucleus of the stria terminalis, lateral division, intermediate part
BSTLP	bed nucleus of the stria terminalis, lateral division, posterior part
BSTLV	bed nucleus of the stria terminalis, lateral division, ventral part
BSTMA	bed nucleus of the stria terminalis, medial division, anterior part
BSTMPI	bed nucleus of the stria terminalis, medial division, posterointermediate part
BSTMPL	bed nucleus of the stria terminalis, medial division, posterolateral part
BSTMPM	bed nucleus of the stria terminalis, medial division, posteromedial part
BSTS	bed nucleus of stria terminalis, supracapsular part
CPu	caudate putamen (striatum)
HDB	nucleus of the horizontal limb of the diagonal band
IPACL	interstitial nucleus of the posterior limb of the anterior commissure, lateral part
IPACM	interstitial nucleus of the posterior limb of the anterior commissure, medial part
LPO	lateral preoptic area
LSD	lateral septal nucleus, dorsal part
LSI	lateral septal nucleus, intermediate part
LSV	lateral septal nucleus, ventral part
MCPO	magnocellular preoptic nucleus
MnPO	median preoptic nucleus
MPOC	medial preoptic nucleus, central part
MPOL	medial preoptic nucleus, lateral part
MPOM	medial preoptic nucleus, medial part
MS	medial septal nucleus
SFi	septofimbrial nucleus
SI	substantia innominata
VLPO	ventrolateral preoptic nucleus
VMPO	ventromedial preoptic nucleus
VOLT	vascular organ of the lamina terminalis
VP	ventral pallidum
ZL	zona limitans
Hippocampus	
CA1	field ca1 of hippocampus
CA3	field CA3 of hippocampus
df	dorsal fornix
DG	dentate gyrus

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dhc	dorsal hippocampal commissure
f	fornix
fi	fimbria of the hippocampus
GrDG	granular layer of the dentate gyrus
LMol	lacunosum moleculare layer of the hippocampus
Mol	molecular layer of the dentate gyrus
Or	oriens layer of the hippocampus
PoDG	polymorph layer of the dentate gyrus
PrS	presubiculum
Py	pyramidal tract
S	subiculum
TS	triangular septal nucleus
Thalamus	
AD	anterodorsal thalamic nucleus
AM	anteromedial thalamic nucleus
APT	anterior pretectal nucleus
APTD	anterior pretectal nucleus, dorsal part
APTV	anterior pretectal nucleus, ventral part
CL	centrolateral thalamic nucleus
CM	central medial thalamic nucleus
DLG	dorsal lateral geniculate nucleus
eml	external medullary lamina
Eth	ethmoid thalamic nucleus
F	nucleus of the fields of Forel
FF	fields of Forel
fr	fasciculus retroflexus
Gus	gustatory thalamic nucleus
hbc	habenular commissure
IAD	interanterodorsal thalamic nucleus
IAM	interanteromedial thalamic nucleus
ic	internal capsule
IGL	intergeniculate leaf
IMD	intermediodorsal thalamic nucleus
LDDM	laterodorsal thalamic nucleus, dorsomedial part
LDVL	laterodorsal thalamic nucleus, ventrolateral part
LGP	lateral globus pallidus
LHb	lateral habenular nucleus
LHbL	lateral habenular nucleus, lateral part
LHbM	lateral habenular nucleus, medial part
LPLR	lateral posterior thalamic nucleus, laterorostral part
LPMC	lateral posterior thalamic nucleus, mediocaudal part
LPMR	lateral posterior thalamic nucleus, mediorostral part
MD	mediodorsal thalamic nucleus
MDC	mediodorsal thalamic nucleus, central part
MDL	mediodorsal thalamic nucleus, lateral part
MDM	mediodorsal thalamic nucleus, medial part
MGD	medial geniculate nucleus, dorsal part
MGM	medial geniculate nucleus, medial part
MGP	medial globus pallidus (entopeduncular nucleus)
MGV	medial geniculate nucleus, ventral part
MHb	medial habenular nucleus

MZMG	marginal zone of the medial geniculate
OPC	oval paracentral thalamic nucleus
OPT	olivary pretectal nucleus
PC	paracentral thalamic nucleus
PF	parafascicular thalamic nucleus
PIL	posterior intralaminar thalamic nucleus
Po	posterior thalamic nuclear group
PoMn	posteromedian thalamic nucleus
PP	peripeduncular nucleus
PPT	posterior pretectal nucleus
PR	prerubral field
PrC	precommissural nucleus
pv	periventricular fiber system
PV	paraventricular thalamic nucleus
PVA	paraventricular thalamic nucleus, anterior part
PVP	paraventricular thalamic nucleus, posterior part
Re	reuniens thalamic nucleus
REth	retroethmoid nucleus
Rh	rhomboid thalamic nucleus
RI	rostral interstitial nucleus of medial longitudinal fasciculus
Rt	reticular thalamic nucleus
SCO	subcommissural organ
SG	suprageniculate thalamic nucleus
sm	stria medullaris of the thalamus
SPF	subparafascicular thalamic nucleus
STh	subthalamic nucleus
Sub	submedius thalamic nucleus
SubG	subgeniculate nucleus
VL	ventrolateral thalamic nucleus
VLGMC	ventral lateral geniculate nucleus, magnocellular part
VLGPC	ventral lateral geniculate nucleus, parvicellular part
VM	ventromedial thalamic nucleus
VPL	ventral posterolateral thalamic nucleus
VPM	ventral posteromedial thalamic nucleus
VRe	ventral reuniens thalamic nucleus
Xi	xiphoid thalamic nucleus
Hypothalamus	
AAV	anterior amygdaloid area, ventral part
ACo	anterior cortical amygdaloid nucleus
AHA	anterior hypothalamic area, anterior part
AHC	anterior hypothalamic area, central part
AHiAL	amygdalohippocampal area, anterolateral part
AHiPM	amygdalohippocampal area, posteromedial part
AHP	anterior hypothalamic area, posterior part
APir	amygdalopiriform transition area
Arc	arcuate hypothalamic nucleus
ArcL	arcuate hypothalamic nucleus, lateral part
ArcLP	arcuate hypothalamic nucleus, lateroposterior part
ArcMP	arcuate hypothalamic nucleus, medial posterior part
AStr	amygdalostriatal transition area
BLA	basolateral amygdaloid nucleus, anterior part

BLP	basolateral amygdaloid nucleus, posterior part
BLV	basolateral amygdaloid nucleus, ventral part
BMA	basolateral amygdaloid nucleus, anterior part
BMP	basomedial amygdaloid nucleus, posterior part
CeC	central amygdaloid nucleus, capsular part
CeL	central amygdaloid nucleus, lateral division
CeM	central amygdaloid nucleus, medial division
CeMAD	central amygdaloid nucleus, medial division, anterodorsal part
CeMPV	central amygdaloid nucleus, medial posteroventral part
cp	cerebral peduncle, basal part
CxA	cortex-amygdala transition zone
DM	dorsomedial hypothalamic nucleus
DMC	dorsomedial hypothalamic nucleus, compact part
DMD	dorsomedial hypothalamic nucleus, dorsal part
DMV	dorsomedial hypothalamic nucleus, ventral part
DTM	dorsal tuberomammillary nucleus
I	intercalated nuclei of the amygdala
IM	intercalated amygdaloid nucleus, main part
IPAC	interstitial nucleus of the posterior limb of the anterior commissure
LA	lateroanterior hypothalamic nucleus
LaDL	lateral amygdaloid nucleus, dorsolateral part
LaVL	lateral amygdaloid nucleus, ventrolateral part
LaVM	lateral amygdaloid nucleus, ventromedial part
LH	lateral hypothalamic area
LM	lateral mammillary nucleus
MCLH	magnocellular nucleus of the lateral hypothalamus
MeA	medial amygdaloid nucleus, anterior part
MeAD	medial amygdaloid nucleus, anteriodorsal part
MePD	medial amygdaloid nucleus, posterodorsal part
MePV	medial amygdaloid nucleus, posteroventral part
ML	medial mammillary nucleus, lateral part
MM	medial mammillary nucleus, medial part
MMn	medial mammillary nucleus, median part
MPA	medial preoptic area
mt	mammillothalamic tract
mtg	mammillotegmental tract
MTu	medial tuberal nucleus
ns	nigrostriatal bundle
opt	optic tract
PaAP	paraventricular hypothalamic nucleus, anterior parvicellular part
PaDC	paraventricular hypothalamic nucleus, dorsal cap
PaLM	paraventricular hypothalamic nucleus, lateral magnocellular part
PaMM	paraventricular hypothalamic nucleus, medial magnocellular part
PaMP	paraventricular hypothalamic nucleus, medial parvicellular part
PaPo	paraventricular hypothalamic nucleus, posterior part
PaV	paraventricular hypothalamic nucleus, ventral part
Pe	periventricular hypothalamic nucleus
PeF	perifornical nucleus
PH	posterior hypothalamic area
PLCo	posterolateral cortical amygdaloid nucleus
PMCo	posteromedial cortical amygdaloid nucleus (C3)

PMD	premammillary nucleus, dorsal part
PMV	premammillary nucleus, ventral part
PSTh	parasubthalamic nucleus
RCh	retrochiasmatic area
SCh	suprachiasmatic nucleus
SChDM	suprachiasmatic nucleus, dorsomedial part
SChVL	suprachiasmatic nucleus, ventrolateral part
SHy	septohypothalamic nucleus
SMT	submammillothalamic nucleus
SO	supraoptic nucleus
SOR	supraoptic nucleus, retrochiasmatic part
sox	supraoptic decussation
SPa	subparaventricular zone of the hypothalamus
Stg	stigmoid hypothalamic nucleus
SubI	subincertal nucleus
SuM	supramammillary nucleus
SuML	supramammillary nucleus, lateral part
SuMM	supramammillary nucleus, medial part
sumx	supramammillary decussation
TC	tuber cinereum area
Te	terete hypothalamic nucleus
VMH	ventromedial hypothalamic nucleus
VMHC	ventromedial hypothalamic nucleus, central part
VMHDM	ventromedial hypothalamic nucleus, dorsomedial part
VMHVL	ventromedial hypothalamic nucleus, ventrolateral part
ZI	zona incerta
ZID	zona incerta, dorsal part
ZIV	zona incerta, ventral part
Midbrain and pons	
3N	oculomotor nucleus
3PC	oculomotor nucleus, parvicellular part
ATg	anterior tegmental nucleus
bic	brachium of the inferior colliculus
BIC	nucleus of the brachium of the inferior colliculus
CGA	central gray, alpha part
CGPn	central gray of the pons
CIC	central nucleus of the inferior colliculus
CnF	cuneiform nucleus
csc	commissure of the superior colliculus
DCIC	dorsal cortex of the inferior colliculus
Dk	nucleus of Darkschewitsch
DLPAG	dorsolateral periaqueductal gray
DMPAG	dorsomedial periaqueductal gray
DMPn	dorsomedial pontine nucleus
DMTg	dorsomedial tegmental area
DpG	deep gray layer of the superior colliculus
DpMe	deep mesencephalic nucleus
DpWh	deep white layer of the superior colliculus
DRC	dorsal raphe nucleus, caudal part
DRI	dorsal raphe nucleus, interfascicular part
DRV	dorsal raphe nucleus, ventral part

DRVL	dorsal raphe nucleus, ventrolateral part
DTgC	dorsal tegmental nucleus, central part
DTgP	dorsal tegmental nucleus, pericentral part
ECIC	external cortex of the inferior colliculus
EMi	epimicrocellular nucleus
EW	Edinger-Westphal nucleus
IF	interfascicular nucleus
InC	interstitial nucleus of Cajal
InCG	interstitial nucleus of Cajal, greater part
InCo	intercollicular nucleus
InG	intermediate gray layer of the superior colliculus
InWh	intermediate white layer of the superior colliculus
IPC	interpeduncular nucleus, caudal subnucleus
IPDL	interpeduncular nucleus, dorsolateral subnucleus
IPDM	interpeduncular nucleus, dorsomedial subnucleus
IPF	interpeduncular fossa
IPI	interpeduncular nucleus, intermediate subnucleus
IPL	interpeduncular nucleus, lateral subnucleus
IPR	interpeduncular nucleus, rostral subnucleus
KF	Ko¨lliker-Fuse nucleus
LC	locus coeruleus
LDTg	laterodorsal tegmental nucleus
LDTgV	laterodorsal tegmental nucleus, ventral part
lfp	longitudinal fasciculus of the pons
LPAG	lateral periaqueductal gray
LPBC	lateral parabrachial nucleus, central part
LPBS	lateral parabrachial nucleus, superior part
LPBV	lateral parabrachial nucleus, ventral part
MA3	medial accessory oculomotor nucleus
MCPC	magnocellular nucleus of the posterior commissure
MiTg	microcellular tegmental nucleus
ml	medial lemniscus
mlf	medial longitudinal fasciculus
MnR	median raphe nucleus
Mo5	motor trigeminal nucleus
mp	mammillary peduncle
MPB	medial parabrachial nucleus
MT	medial terminal nucleus of the accessory optic tract
Op	optic nerve layer of the superior colliculus
Pa4	paratrochlear nucleus
PAG	periaqueductal gray
PBP	parabrachial pigmented nucleus
PCom	nucleus of the posterior commissure
PMnR	paramedian raphe nucleus
Pn	pontine nuclei
PN	paranigral nucleus
PnC	pontine reticular nucleus, caudal part
PnO	pontine reticular nucleus, oral part
PnV	pontine reticular nucleus, ventral part
PPTg	pedunculopontine tegmental nucleus
RC	raphe cap

RLi	rostral linear nucleus of the raphe
RMC	red nucleus, magnocellular part
RPC	red nucleus, parvicellular part
RPF	retroparafascicular nucleus
RPO	rostral periolivary region
RRF	retrorubral field
RtTg	reticulotegmental nucleus of the pons
RtTgP	reticulotegmental nucleus of the pons, pericentral part
Sag	sagulum nucleus
SC	superior colliculus
scp	superior cerebellar peduncle (brachium conjunctivum)
SNC	substantia nigra, compact part
SNL	substantia nigra, lateral part
SNR	substantia nigra, reticular part
Su3	supraoculomotor periaqueductal gray
Su3C	supraoculomotor cap
SubB	subbrachial nucleus
SuG	superficial gray layer of the superior colliculus
ts	tectospinal tract
VLPAG	ventrolateral periaqueductal gray
VLTg	ventrolateral tegmental area
VTA	ventral tegmental area
vtgx	ventral tegmental decussation
VTRZ	visual tegmental relay zone
xscp	decussation of the superior cerebellar peduncle
Zo	zonal layer of the superior colliculus
Medulla	
7N	facial nucleus
10N	dorsal motor nucleus of vagus
12N	hypoglossal nucleus
Amb	ambiguus nucleus
AP	area postrema
CeCv	central cervical nucleus
cu	cuneate fasciculus
Cu	cuneate nucleus
DMSp5	dorsomedial spinal trigeminal nucleus
DPGi	dorsal paragigantocellular nucleus
DPO	dorsal periolivary region
ECu	external cuneate nucleus
EVe	nucleus of origin of efferents of the vestibular nerve
FVe	F cell group of the vestibular complex
Gi	gigantocellular reticular nucleus
GiA	gigantocellular reticular nucleus, alpha part
GiV	gigantocellular reticular nucleus, ventral part
Gr	gracile nucleus
icp	inferior cerebellar peduncle (restiform body)
In	intercalated nucleus of the medulla
InM	intermedius nucleus of the medulla
IOB	inferior olive, subnucleus B of medial nucleus
IOBe	inferior olive, subnucleus B of medial nucleus
IOC	inferior olive, subnucleus C of medial nucleus

IOD	inferior olive, dorsal nucleus
IOK	inferior olive, cap of Kooy of the medial nucleus
IOVL	inferior olive, ventrolateral protrusion
IRt	intermediate reticular nucleus
Li	linear nucleus of the medulla
LPGi	lateral paragigantocellular nucleus
LRt	lateral reticular nucleus
LRtPC	lateral reticular nucleus, parvicellular part
LSO	lateral superior olive
LVe	lateral vestibular nucleus
LVPO	lateroventral periolivary nucleus
MdD	medullary reticular nucleus, dorsal part
MdV	medullary reticular nucleus, ventral part
MVe	medial vestibular nucleus
MVeMC	medial vestibular nucleus, magnocellular part
MVePC	medial vestibular nucleus, parvicellular part
PCRt	parvicellular reticular nucleus
PCRtA	parvicellular reticular nucleus, alpha part
PMn	paramedian reticular nucleus
PPy	parapyramidal nucleus
Pr	prepositus nucleus
Pr5DM	principal sensory trigeminal nucleus, dorsomedial part
Pr5VL	principal sensory trigeminal nucleus, ventrolateral part
PSol	parasolitary nucleus
py	pyramidal tract
pyx	pyramidal decussation
RAmb	retroambiguus nucleus
RMg	raphe magnus nucleus
Ro	nucleus of Roller
ROb	raphe obscurus nucleus
RPa	raphe pallidus nucleus
RVL	rostroventrolateral reticular nucleus
Sol	nucleus of the solitary tract
SolC	nucleus of the solitary tract, commissural part
SolCe	nucleus of the solitary tract, central part
SolDL	solitary nucleus, dorsolateral part
SolDM	nucleus of the solitary tract, dorsomedial part
SolG	nucleus of the solitary tract, gelatinous part
SolI	nucleus of the solitary tract, interstitial part
SolIM	nucleus of the solitary tract, intermediate part
SolM	nucleus of the solitary tract, medial part
SolV	solitary nucleus, ventral part
SolVL	nucleus of the solitary tract, ventrolateral part
sp5	spinal trigeminal tract
Sp5C	spinal trigeminal nucleus, caudal part
Sp5I	spinal trigeminal nucleus, interpolar part
Sp5O	spinal trigeminal nucleus, oral part
Sp5ODM	spinal trigeminal nucleus, oral part, dorsomedial division
Sp5OVL	spinal trigeminal nucleus, oral part, ventrolateral division
SPO	superior paraolivary nucleus
SpVe	spinal vestibular nucleus

SuVe	superior vestibular nucleus
VCA	ventral cochlear nucleus, anterior part
vsc	ventral spinocerebellar tract
X	nucleus X
Cerebellum	
1Cb	1st Cerebellar lobule
2Cb	2nd Cerebellar lobule
3Cb	3rd Cerebellar lobule
4&5Cb	4&5th Cerebellar lobules
6Cb	6th Cerebellar lobule
7Cb	7th Cerebellar lobule
8Cb	8th Cerebellar lobule
9Cb	9th Cerebellar lobule
10Cb	10th Cerebellar lobule
Crus1	crus 1 of the ansiform lobule
Crus2	crus 2 of the ansiform lobule
Fl	flocculus
PFl	paraflocculus
PM	paramedian lobule
Sim	simple lobule
Ventricular zones	
3V	3rd ventricle
OV	olfactory ventricle (olfactory part of lateral ventricle)
OV	olfactory ventricle (olfactory part of lateral ventricle)
SVZ	subventricular zone

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Author notes

Competing interests: M.Z. is inventor on issued and pending patents on the use of FSH as a target for osteoporosis, obesity and Alzheimer’s disease. M.Z. and T.Y. are inventors on a pending patent on an FSH antibody that is formulated at ultrahigh concentration. The patents will be held by Icahn

School of Medicine at Mount Sinai, and M.Z. and T.Y. would be recipients of royalties, *per* institutional policy. The other authors declare no competing financial interests.

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Peer reviews

Reviewer #1 (Public review):

Summary:

Despite accumulating prior studies on the expressions of AVP and AVPR1a in the brain, a detailed, gender-specific mapping of AVP/AVPR1a neuronal nodes has been lacking. Using RNAscope, a cutting-edge technology that detects single RNA transcripts, the authors created a comprehensive neuroanatomical atlas of Avp and Avpr1a in male and female brains.

Strengths:

This well-executed study provides valuable new insights into gender differences in the distribution of Avp and Avpr1a. The atlas is an important resource for the neuroscience community.

The authors have adequately addressed all of my concerns. I have no further questions or concerns.

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Reviewer #2 (Public review):

Summary:

The authors conducted a brain-wide survey of Avp (arginine vasopressin) and its Avpr1a gene expression in the mouse brain using RNAscope, a high-resolution in situ hybridization method. Overall, the findings are useful and important because they identify brain regions that express the Avpr1a transcript. A comprehensive overview of Avpr1a expression in the mouse brain could be highly informative and impactful. The authors used RNAscope (a proprietary in situ hybridization method) to assess transcript abundance of Avp and one of its receptors, Avpr1a. The finding of Avp-expressing cells outside the hypothalamus and the extended amygdala is novel and is nicely demonstrated by new photomicrographs in the revised manuscript. The Avpr1a data suggest expression in numerous brain regions. In the revised manuscript, reworked figures make the data easier to interpret.

Strengths:

A survey of Avpr1a expression in the mouse brain is an important tool for exploring vasopressin function in the mammalian brain and for developing hypotheses about cell- and circuit-level function.

Future considerations:

The work contained in the manuscript is substantial and informative. Some questions remain and would be addressed in the current manuscript. How many cells are impacted? Are transcripts spread across many cells or only present in a few cells? Is density evenly distributed through a brain region or compacted into a subfield?

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Author response:

The following is the authors' response to the original reviews.

> ***Reviewer #1:***

We thank the reviewer for great suggestions.

> *(1) The X-axis labels in some panels in Figure 2C and Supplementary Figure 2B overlap, making them difficult to read. Adjusting the label spacing or formatting would improve clarity.*

We thank the reviewer for the comment. All panels including Figure 2C and Supplementary Figure 2B, have now been organized the way in which X-axis labels are easily read.

> *(2) In the scatter dot plot bar diagrams, it appears that n=3 for most of the data. Does this represent the number of mice used or the number of tissue sections per sample? This should be clarified in the figure legends for better transparency.*

Great suggestion. In Results (page 7, lines 135-136), we now clarified that quantification was performed on every tenth section of the brain from 3 female and 3 male mice. Additionally, in the legends for scatter dot plot bar diagrams we also mentioned that n=3 represents the number of mice used.

> *(3) In Supplemental Figure 2B, the positive signals are not clearly visible. Providing higher-magnification images is recommended.*

Great suggestion. The revised Supplemental Figure 2B, but also Figure 2A, now provide higher magnification inset images with distinctive positive signals.

> ***Reviewer #2:***

We thank the reviewer for great and critical suggestions.

> *(1) Introduction:*
>
> *Line 58: References should be provided for this statement as it is based on a robust field of research, not on a new concept.*

We thank the reviewer for the comment. We have now included relevant references as suggested (page 4, line 58).

> *(2) Line 100-102: This sentence seems to make new, an idea that has been well-documented since the late 1970s. Posterior pituitary hormones oxytocin and vasopressin have long been known to have multiple peripheral targets, and at least a subset of vasopressin and oxytocin neurons have robust central projections. The central targets have been the focus of study for numerous labs. Reference 34 does not relate to posterior pituitary hormones and seems mis-cited.*

We have changed this sentence, excluded the reference that does not relate to posterior pituitary hormones and added 4 further references reporting other non-traditional roles of vasopressin and oxytocin (page 6, lines 100-102).

> *(3) Lines 102-108: While the regulation of bone is an interesting example of an under-appreciated impact of vasopressin, the example does not build to the rationale for examining central Avp and Avpr1a expression.*

We mean no disrespect here, but we have recently reported neural brain-bone connections using the SNS-specific PRV152 virus (Ryu et al., 2024; PMID: 38963696) and submitted Single Transcript Level Atlas of Oxytocin and the Oxytocin Receptor in the Mouse Brain (doi: *https://doi.org/10.1101/2024.02.15.580498*). Surprisingly, we detected *Avpr1a* and *Oxtr* expression in certain brain areas (for example, PVH and MPOM) that connect to both bone and adipose tissue through the SNS—raising an important question regarding a central role of *Avpr1a* and *Oxtr* in bodily mass and fat regulation.

> *(4) Line 111: Avp expression and Avpr1a expression have both been studied using in situ hybridization. Thus, the overall concept is less novel than hinted at in the text. Avp expression has been studied quite extensively. Avpr1a expression has not been studied in an exhaustive fashion.*

We thank the reviewer for this comment and absolutely agree that brain AVP expression has been studied extensively. As with the *Avpr*, we believe that RNAscope probe design and signal amplification system employed in our study allow for more specific and sensitive detection of individual RNA targets at the single transcript level with much cleaner background noise comparing to *in situ* hybridization method.

> *(5) Results:*
>
> *Line 143: RNAscope is indeed a powerful method of detecting mRNA at the single transcript level. However, using that single transcript resolution only to provide transcript per brain region analysis, losing all of the nuance of the individual transcript expression, seems like a poor use of the method potential.*

This is a good point and we did notice that *Avpr1a* transcript expression in several brain nuclei displayed individual pattern of expression versus more ubiquitous expression in most of the other brain areas. We noted this finding in Results (page 9, lines 164-168); however, because of the word limits in Discussion, we are not sure what would be dropped to make more room and whether this is truly necessary.

> *(6 &7) Line 135: Sections were coded from 3 males and 3 females. I would argue that there is not enough statistical power to make inferences regarding sex differences or regional differences. In fact, the authors did not provide any statistical analysis in the manuscript at all, even though they stated they had completed statistical tests on the methods.*
>
> *150-157: All statements regarding sex differences in expression are made without statistical analyses, which, if conducted, would be underpowered. Given the limitations of performing and analyzing RNAscope data en masse a low n is understandable, but it requires a much more precise description of the data and a more careful look at how the results can be interpreted.*

We thank the reviewer for these comments. We mean no disrespect here, but while statistical analysis for main brain regions is relevant, it is not meaningful as far as nuclei, sub-nuclei and regions are concerned. It is noteworthy to mention that RNAscope data analysis in the whole mouse brain is an extremely drawn-out process requiring almost 2 months to conduct exhaustive manual counting of single *Avpr1a* transcripts in a single mouse brain—authors analyzed 6 brains. That said, statistical tests have been performed and exact *P* values are now shown in graphs.

> *(8) Line 146: I am flagging this instance, but it should be corrected everywhere it occurs. Since we cannot know the gender of a given mouse, I would recommend referring to the mouse's "sex" rather than its "gender."*

Good suggestion. We made adequate changes throughout the manuscript.

> *(9) Line 153: The authors switch to discussing cell numbers. Why is this data relegated to the supplemental material?*

Main figures in the manuscript report *Avp* and *Avpr1a* transcript density which has more important biological significance in terms of signal efficiency and cellular response dynamics. Due to the graph abundancy in the main text, we included all graphs with *Avp* and *Avpr1a* transcript counts in the supplemental material.

> *(10) Methods:*
>
> *Line 369: "For simplicity and clarity, exact test results and exact P values are not presented." Simplicity or clarity is not a scientific rationale not to provide accurate statistics.*

We now provide exact *P* values in the graphs and the sentence in line 369 has been corrected accordingly (page 18, lines 379-380).

> *(11) Line 362: The description of how data were analyzed is inadequate. More detail is needed.*

Agreed. We now included a detailed description on how data was analyzed (page 18, lines 365-374).

> *(12) Discussion:*
>
> *Line 321: "This contrasts the rudimentary attribution of a single function per brain area." While brain function is often taught in such rudimentary terms to make the information palatable to students, I do not think the scientific literature on vasopressin function published over the past 50 years would suggest that we are so naïve in interpreting the functional role of vasopressin in the brain. Clearly, vasopressin is involved in numerous brain functions that likely cross behavioral modalities.*

Agreed and we removed this sentence.

> *(13) Line 322: "The approach of direct mapping of receptor expression in the brain and periphery provides the groundwork." On its face, this statement is true, but the present data build on the groundwork laid by others (multiple papers from Ostrowski et al. in the early 1990s).*

Agreed.

> *(14) Figures:*

Figure 1: 1B, I do not know the purpose of creating graphs with single bars (3V, ic, pir-male, and pir-female); there are no comparisons made in the graph. In the graphs with many brain regions, very little data can be effectively represented with the scale as it is. I recommend using tables to provide the count/density data and making graphs of only the most robust areas. In addition, although there is no statistical comparison, combining males and females in the same graphs might be beneficial to make a visual comparison easier. Why were cell counts only included in the supplemental material? Is that data not relevant?

We thank the reviewer for this comment. Now all figures are presented in a more effective and aesthetically pleasing way.

> *(15) There is a real missed opportunity to highlight some of the findings. For example, cell counts and density measures are provided for regions in the hippocampus, thalamus, and cortex that are not typically reported to contain vasopressin-expressing cells. Photomicrographs of these locations showing the RNAscope staining would be far more impactful in reporting these data. Are there cells expressing Avp, or is the Avp mRNA in these areas contained in fibers projecting to these areas from hypothalamic and forebrain sources?*

Great suggestion. We now see in Figure 1D showing novel *Avp* transcript expression in the hippocampus, thalamus and cortex. Based on counterstained hematoxylin staining, *Avp* mRNA transcripts were found in somata.

> *(16) Figure 1C legend suggests images of the hippocampus and cortex, but all images are from the hypothalamus. Abbreviations are not defined.*

Thank you for the comment. We corrected Figure 1C legend and separately included Figure 1D showing novel Avp mRNA expression in the hippocampus and cortex.

> *(17) Figure 2: The analysis of Avpr1a suffers from some of the same issues as the Avp analysis. In Figure 2A, the photomicrographs do not do a very good job of illustrating representative staining. The central canal image does not appear to have any obvious puncta, but the density of Avpr1a puncta suggests something different. The sex difference in the arcuate is also not clearly apparent in representative images. There is minimal visualization of the data for a project that depends so heavily on the appearance of puncta in tissue, coupled with the lack of clarity in the images provided, greatly diminished the overall enthusiasm for the data presentation. The figures in 2C would be more useful as tables with graphs used to highlight specific regions; as is, most of the data points are lost against the graph axis. Photomicrographs would also provide a better understanding of the data than graphs.*

Great suggestion. The revised Figure 2A but also Supplemental Figure 2B now provide higher magnification inset images with distinctive positive signals. As with Figures 2C, we arranged all graphs in a more effective and aesthetically pleasing manner.

> *(18) Figure 3: Given the low number of animals and, therefore, low statistical power, I do not think that illustrating the ratios of male to female is a statistically valid comparison.*

Please see response to Point 6 & Point 7.

> *(19) Figure 4: Pituitary is an interesting choice to analyze. However, why was only the posterior pituitary analyzed? Were Avp transcripts contained in terminals of vasopressin neuron axons or other cells? Was Avpr1a transcript present in blood vessel cells where Avp is released? A different cell type? Why not examine the anterior pituitary, which also expresses Avp receptors (although the literature suggests largely Avpr1b)?*

Thank you for the great comment. We included only posterior pituitary because there were no positive *Avp/Avpr1a* transcripts found in the anterior pituitary. Unfortunately, we have not performed cell type-specific staining, which would have enabled greater variation in AVP and its receptor expression across various cell types.

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